

5G RAN

Massive MIMO AHR (Low-Frequency TDD) Feature Parameter Description

Issue	Draft B
Date	2026-01-30



Copyright © Huawei Technologies Co., Ltd. 2026. All rights reserved.

No part of this document may be reproduced or transmitted in any form or by any means without prior written consent of Huawei Technologies Co., Ltd.

Trademarks and Permissions



HUAWEI and other Huawei trademarks are trademarks of Huawei Technologies Co., Ltd.

All other trademarks and trade names mentioned in this document are the property of their respective holders.

Notice

The purchased products, services and features are stipulated by the contract made between Huawei and the customer. All or part of the products, services and features described in this document may not be within the purchase scope or the usage scope. Unless otherwise specified in the contract, all statements, information, and recommendations in this document are provided "AS IS" without warranties, guarantees or representations of any kind, either express or implied.

The information in this document is subject to change without notice. Every effort has been made in the preparation of this document to ensure accuracy of the contents, but all statements, information, and recommendations in this document do not constitute a warranty of any kind, express or implied.

Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

Security Declaration

Vulnerability

Huawei's regulations on product vulnerability management are subject to the *Vul. Response Process*. For details about this process, visit the following web page:

<https://www.huawei.com/en/psirt/vul-response-process>

For vulnerability information, enterprise customers can visit the following web page:

<https://securitybulletin.huawei.com/enterprise/en/security-advisory>

Contents

1 Change History.....**1**

2 About This Document.....**3**

2.1 General Statements..... 3

2.2 Features in This Document..... 3

2.3 Differences..... 4

3 Overview.....**5**

4 AHR Introduction.....**6**

4.1 Principles..... 6

4.1.1 Adaptive CSI-RS Multi-Beam Adjustment..... 6

4.1.2 Adaptive Multi-Stream Perception..... 7

4.2 Network Analysis..... 8

4.2.1 Benefits..... 9

4.2.2 Impacts..... 9

4.3 Requirements..... 10

4.3.1 Licenses..... 10

4.3.2 Functions..... 11

4.3.2.1 Prerequisite Functions..... 11

4.3.2.2 Mutually Exclusive Functions..... 12

4.3.2.3 Function Impacts..... 13

4.3.3 Hardware..... 15

4.3.4 Cells..... 15

4.4 Operation and Maintenance..... 15

4.4.1 Data Configuration..... 15

4.4.1.1 Data Preparation..... 16

4.4.1.2 Using MML Commands..... 17

4.4.1.3 Using the MAE-Deployment..... 18

4.4.2 Activation Verification..... 18

4.4.3 Network Monitoring..... 18

5 AHR Experience Turbo 2.0.....**20**

5.1 Principles..... 20

5.1.1 SRS Interference Coordination..... 20

5.1.2 Precise AMC..... 22

5.2 Network Analysis.....	25
5.2.1 Benefits.....	25
5.2.2 Impacts.....	26
5.3 Requirements.....	27
5.3.1 Licenses.....	28
5.3.2 Functions.....	28
5.3.2.1 Prerequisite Functions.....	28
5.3.2.2 Mutually Exclusive Functions.....	29
5.3.2.3 Function Impacts.....	30
5.3.3 Hardware.....	31
5.3.4 Cells.....	32
5.3.5 Networking.....	34
5.3.6 Others.....	34
5.4 Operation and Maintenance.....	34
5.4.1 Data Configuration.....	34
5.4.1.1 Data Preparation.....	34
5.4.1.2 Using MML Commands.....	37
5.4.1.3 Using the MAE-Deployment.....	38
5.4.2 Activation Verification.....	38
5.4.3 Network Monitoring.....	39
6 AHR Capacity Upgrade 2.0.....	40
6.1 Principles.....	40
6.1.1 High-Resolution MU Pairing.....	40
6.1.2 Adaptive MU Resource Management.....	42
6.2 Network Analysis.....	44
6.2.1 Benefits.....	44
6.2.2 Impacts.....	45
6.3 Requirements.....	46
6.3.1 Licenses.....	46
6.3.2 Functions.....	47
6.3.2.1 Prerequisite Functions.....	47
6.3.2.2 Mutually Exclusive Functions.....	47
6.3.2.3 Function Impacts.....	49
6.3.3 Hardware.....	51
6.3.4 Cells.....	53
6.3.5 Others.....	54
6.4 Operation and Maintenance.....	54
6.4.1 Data Configuration.....	54
6.4.1.1 Data Preparation.....	54
6.4.1.2 Using MML Commands.....	56
6.4.1.3 Using the MAE-Deployment.....	57
6.4.2 Activation Verification.....	57

6.4.3 Network Monitoring.....57

7 Parameters..... 59

8 Counters..... 61

9 Glossary..... 62

10 Reference Documents..... 63

1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes
None
- Editorial changes
Revised descriptions of the impact relationship between precise RB control for downlink retransmissions and power optimization for downlink retransmissions. For details, see [5.3.2.3 Function Impacts](#).
Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 03 (2025-12-14).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between downlink experience satisfaction guarantee and load-based adaptive downlink scheduling. For details, see 6.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between precise RB control for downlink retransmissions and power optimization for downlink retransmissions. For details, see 5.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

- Editorial changes
Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following NR TDD features.

Feature ID	Feature Name	Chapter/Section
FOFD-051301	AHR Introduction	4 AHR Introduction
FOFD-061201	AHR Experience Turbo 2.0	5 AHR Experience Turbo 2.0
FOFD-061202	AHR Capacity Upgrade 2.0	6 AHR Capacity Upgrade 2.0

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
AHR Introduction	None	4 AHR Introduction
AHR Experience Turbo 2.0	None	5 AHR Experience Turbo 2.0
AHR Capacity Upgrade 2.0	None	6 AHR Capacity Upgrade 2.0

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
AHR Introduction	Supported only in low frequency bands	4 AHR Introduction
AHR Experience Turbo 2.0	Supported only in low frequency bands	5 AHR Experience Turbo 2.0
AHR Capacity Upgrade 2.0	Supported only in low frequency bands	6 AHR Capacity Upgrade 2.0

3 Overview

The Adaptive High Resolution (AHR) solution aims to tackle the challenges arising from various types of UEs, user service behaviors, and interference when multiple cells are configured for contiguous coverage in TDD massive MIMO scenarios. AHR improves user experience and capacity in massive MIMO cells by focusing on channel estimation, weight adjustment, and UE pairing performance improvement and by enabling link adaptation and adaptive service scheduling policies. The AHR solution comprises the following features:

- **AHR Introduction:** It provides adaptive CSI-RS multi-beam adjustment and adaptive multi-stream perception. The former function enhances CSI-RS coverage by flexibly and dynamically configuring the number of CSI-RS resources. The latter function selects optimal UEs and scheduling schemes for MU-MIMO pairing to shorten the service delay for UEs and increase the user-perceived rate. (CSI-RS is short for channel state information reference signal.)
- **AHR Experience Turbo 2.0:** It provides SRS interference coordination and precise AMC. The former function reduces SRS interference and improves SRS quality and weight accuracy, thereby increasing the downlink user-perceived rate. The latter function improves the accuracy of MCS indexes and optimizes MIMO pairing performance, thereby improving the spectral efficiency and increasing the downlink user-perceived rate. (SRS is short for sounding reference signal and MCS is short for modulation and coding scheme.)
- **AHR Capacity Upgrade 2.0:** It provides high-resolution MU pairing and adaptive MU resource management. The former function allows more suitable UEs to participate in MU pairing and increases the numbers of paired UEs and PRBs. The latter function increases MU-MIMO resource usage and optimizes resource allocation. This enhances the overall MU pairing performance and further improves cell capacity and user experience.

4 AHR Introduction

4.1 Principles

AHR Introduction provides the following functions:

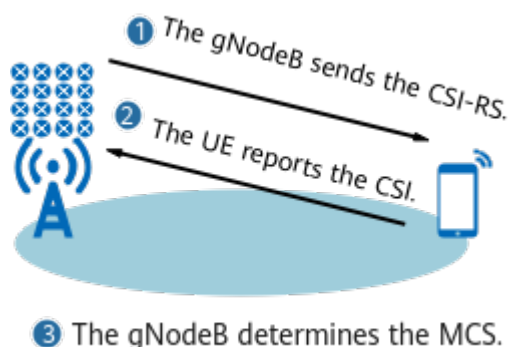
- **Adaptive CSI-RS multi-beam adjustment** flexibly and dynamically configures the number of CSI-RS resources, improving the CSI-RS coverage.
- **Adaptive multi-stream perception** selects optimal UEs and scheduling schemes for MU-MIMO pairing to shorten the service delay for UEs and increase the user-perceived rate.

This feature is controlled by the **AHR_PHASE1_SW** option of the **NRDUCellFeatureSw.AhrSwitch** parameter.

4.1.1 Adaptive CSI-RS Multi-Beam Adjustment

For TDD massive MIMO cells, the CSI-RS can be applied to multiple beam types, which are specified by the **NRDUCellCsirs.CsirsBeamType** parameter. The base station sends the CSI-RS through configured beams. Then, based on the UE's CSI-RS measurement and CSI reporting, the base station selects an optimal MCS for the UE. [Figure 4-1](#) shows this process.

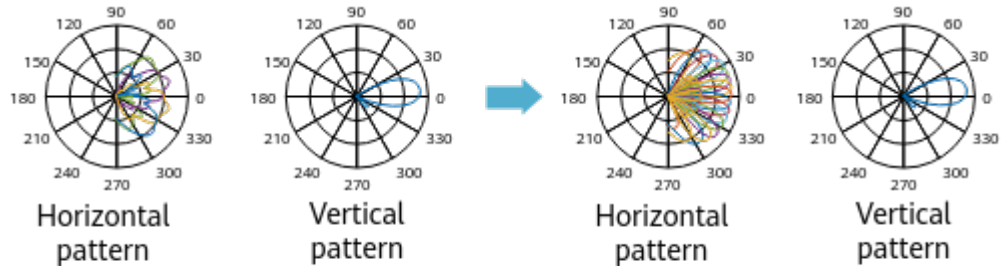
Figure 4-1 CSI reporting and MCS selection



Adaptive CSI-RS multi-beam adjustment flexibly and dynamically configures the number of CSI-RS resources and supports more CSI-RS beams, thereby improving CSI-RS coverage. In SU-MIMO and MU-MIMO scenarios, different beams are adaptively used to reduce interference between data streams.

Figure 4-2 shows the coverage improvement of CSI-RS beam densification.

Figure 4-2 Coverage improvement of CSI-RS beam densification



The number of horizontal beams increases, leading to greater gains.

This function takes effect when the **NRDUCellCsirs.CsirsCellResourceNum** parameter is set to **FD_RESOURCE**. It also involves the following parameters:

- **NRDUCellCsirs.FdCsirsActiveUeThld** and **NRDUCellCsirs.FdCsirsPathLossThld**, which specify the UE number threshold and path loss threshold, respectively, for adjusting the number of CSI-RS resources
- **NRDUCellCsirs.FdCsiPeriod**, which specifies the period of CSI-RS transmission

4.1.2 Adaptive Multi-Stream Perception

Adaptive multi-stream perception selects optimal UEs and scheduling schemes for MU-MIMO pairing to shorten the service delay for UEs and increase the user-perceived rate. It includes the following sub-functions:

- Resource-based adaptive scheduling for massive MIMO
- Experience-based adaptive scheduling for massive MIMO

Resource-based Adaptive Scheduling for Massive MIMO

On live networks, traffic models differ significantly between UEs. For TDD massive MIMO cells, resource-based adaptive scheduling for massive MIMO selects appropriate UEs for downlink MU pairing based on RBs occupied by UEs, traffic models, and signal quality to minimize their overall transmission delay. This shortens the waiting time for UE buffer data to be scheduled, thus increasing the user-perceived rate.

Resource-based adaptive scheduling for massive MIMO is controlled by the **RES_BASED_MM_ADAPT_SCH_SW** option of the **NRDUCellServExp.ServiceExpAlgoSwitch** parameter. Whether this function takes effect once this option is selected varies depending on the scenario, as described in [Table 4-1](#).

Table 4-1 Activation status of resource-based adaptive scheduling for massive MIMO

Scenario	Function Activation Status	Related Parameter
- The duration threshold for optimized downlink scheduling (threshold 1^a) is not zero. - The duration threshold for adaptive downlink scheduling (threshold 2^b) is zero.	- If the estimated downlink data transmission duration of a UE is less than threshold 1, the downlink scheduling priority of the UE will increase. - If the estimated downlink data transmission duration of a UE is greater than or equal to threshold 1, the downlink scheduling priority of the UE will remain unchanged.	a: Threshold 1 is specified by the NRDUCellPdsch.DLSchOptTimeThld parameter. b: Threshold 2 is specified by the NRDUCellDLSch.DLAdaptSchTimeThld parameter.
- Threshold 1 is not zero. - Threshold 2 is not zero.	- If the estimated downlink data transmission duration of a UE is less than the smaller of either threshold 1 or threshold 2, the downlink scheduling priority of the UE will increase. - If the estimated downlink data transmission duration of a UE is greater than or equal to either threshold 1 or threshold 2, the downlink scheduling priority of the UE will remain unchanged.	
Threshold 1 is zero.	The function does not take effect.	

Experience-based Adaptive Scheduling for Massive MIMO

For TDD massive MIMO cells, experience-based adaptive scheduling for massive MIMO adjusts the pairing priority during downlink MU pairing, adapting it to user experience to increase the average downlink UE throughput. This function is controlled by the **EXP_BASED_MM_ADAPT_SCH_SW** option of the **NRDUCellServExp.ServiceExpAlgoSwitch** parameter. It takes effect for a UE only when the buffer size of the UE exceeds the value of the **NRDUCellPdsch.DLDelaySchBufferThld** parameter.

4.2 Network Analysis

4.2.1 Benefits

When AHR Introduction is in effect, the average downlink UE throughput increases when the load is medium or heavy (for example, when the downlink PRB usage is greater than 30%). The actual benefit depends on factors such as the traffic model and network load. The higher the downlink PRB usage, average number of downlink active UEs, or downlink CCE allocation failure rate, the greater the benefit. However, the benefit decreases when a single traffic model is used (for example, when only large- or small-packet services exist).

For AHR Introduction to achieve its optimal benefits, PDSCH MU-MIMO must also be enabled. Otherwise, AHR Introduction provides fewer or even no benefits. PDSCH MU-MIMO is controlled by the **DL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch.MuMimoSwitch** parameter. For details about PDSCH MU-MIMO, see *MIMO (TDD)*.

In congestion scenarios (for example, when the downlink PRB usage exceeds 95% for an extended period), the benefit of adaptive multi-stream perception decreases, but there are no negative impacts.

For the definitions of related indicators, see [4.4.3 Network Monitoring](#).

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

Adaptive CSI-RS multi-beam adjustment has the following network impacts:

- When the CPU load is high (for example, higher than 60%), the average downlink cell throughput and average downlink UE throughput will decrease.
- Other impacts are described in [Table 4-2](#).

Table 4-2 Network impacts of adaptive CSI-RS multi-beam adjustment

Item	Impact
Number of RRC reconfigurations	Increase
Average CQI	Increase
Number of layers paired for downlink MU-MIMO: N.ChMeas.MIMO.DL.Pair.Layer	Possible increase
Average uplink UE throughput	
Proportion of CQIs with a value of 0	
SgNB Mod Required Succ Rate	Possible decrease

Adaptive multi-stream perception has the following network impacts:

- Resource-based adaptive scheduling for massive MIMO results in impacts that are detailed in [Table 4-3](#).

Table 4-3 Network impacts of resource-based adaptive scheduling for massive MIMO

Item	Impact
DU delay	Increase
Average downlink MCS index	Fluctuation
Average downlink cell throughput	Fluctuation
Downlink PRB usage	Fluctuation
Number of uplink active UEs: N.User.RRCCConn.Active.UL.Avg	Fluctuation
Number of PRBs for downlink MU pairing: N.ChMeas.MIMO.DL.Pair.PRB	Fluctuation
Number of PDCCH CCEs: - Average number of used PDCCH CCEs: N.CCE.Used.Avg - Average number of PDCCH CCEs used for downlink DCI: N.CCE.DLDCI.Used.Avg - Average number of PDCCH CCEs used for uplink DCI: N.CCE.ULDCI.Used.Avg	Fluctuation
Average uplink RSSI of channels: N.UL.RSSI.Avg	Fluctuation

- Experience-based adaptive scheduling for massive MIMO increases first packet delay and is therefore not recommended in scenarios that are sensitive to first packet delay.

For the definitions of related indicators, see [4.4.3 Network Monitoring](#).

4.3 Requirements

4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-051301	AHR Introduction	NR0S00AET100	per Cell

Feature ID	Feature Name	Model	Sales Unit
For the license control item NROS00AET100: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation: – When the license sales number pre-check function is enabled, the AHR_PHASE1_SW option of the NRDUCellFeatureSw.AhrSwitch parameter cannot be modified due to license insufficiency, preventing cell activation failures. – When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. The license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the gNBLicenseAlmThld.LicenseChkSw parameter.			

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	For LampSite base stations, distributed massive MIMO must be enabled.

4.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
PMI-based weight optimization	PMI_WEIGHT_OPT_SW option of the NRDUCellBeamAlgo.WeightAlgoSwitch parameter	<i>MIMO (TDD)</i>	When both adaptive CSI-RS multi-beam adjustment and PMI-based weight optimization are enabled: - For 32T32R cells, the NRDUCellCsirs.FdCsiPeriod parameter cannot be set to SLOT40 or SLOT80. - For 64T64R cells, the NRDUCellCsirs.FdCsiPeriod parameter cannot be set to SLOT40, SLOT80, or SLOT160.
Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	In hyper or combined cells using the 32T32R+64T64R mixed configuration, adaptive CSI-RS multi-beam adjustment cannot be enabled.
Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	
Level-2 low jitter and high reliability	LOW_JIT_HIGH_RELIABILITY_LEVEL2_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	None	Experience-based adaptive scheduling for massive MIMO cannot be enabled together with the level-2 low jitter and high reliability function.
Interference avoidance during PUSCH/PDSCH scheduling	UL_CELL_LEVEL_INTRF_AVOID and DL_UE_LEVEL_INTRF_AVOID options of the NRDUCellIntrfIdent.IntrfOptimizationMethod parameter NRDUCellIntrfIdent.IntrfIdentificationMethod set to UL_DL_MANUAL	<i>LTE and NR Zero Bufferzone</i>	Bi-Carrier must be disabled if adaptive CSI-RS multi-beam adjustment and interference avoidance during PUSCH/PDSCH scheduling are both enabled and there is at least one interference range for which the NRDUCellIntrfBand.InterferenceDirection parameter is set to UPLINK_DOWNLINK .

Function Name	Function Switch	Reference	Description
Bi-Carrier	BI_CARRIER_SW option of the NRDUCellFeatureSw. MultiCarrierSwitch parameter	None	
Bi-Carrier	BI_CARRIER_SW option of the NRDUCellFeatureSw. MultiCarrierSwitch parameter	None	When both adaptive CSI-RS multi-beam adjustment and Bi-Carrier are enabled: - For 32T32R cells, the NRDUCellCsirs.FdCsiPeriod parameter cannot be set to SLOT80. - For 64T64R cells, the NRDUCellCsirs.FdCsiPeriod parameter cannot be set to SLOT160.

4.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
PDCCH uplink-to-downlink CCE ratio adaptation	UL_DL_CCE_RATIO_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoSwitch parameter	<i>Channel Management</i>	When both PDCCH uplink-to-downlink CCE ratio adaptation and resource-based adaptive scheduling for massive MIMO are enabled, the number of uplink active UEs may increase. As a result, the uplink cell throughput decreases when there are uplink services.
Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	For UEs configured with SCells for downlink intra-band CA, the activation of resource-based adaptive scheduling for massive MIMO may be affected by the mechanism of data distribution to SCells.

Function Name	Function Switch	Reference	Description
Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	For UEs configured with SCells for downlink intra-FR inter-band CA, the activation of resource-based adaptive scheduling for massive MIMO may be affected by the mechanism of data distribution to SCells.
Inter-gNodeB CA	INTER_GNODEB_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>CA Experience Improvement</i>	For UEs configured with SCells for inter-gNodeB CA, the activation of resource-based adaptive scheduling for massive MIMO may be affected by the mechanism of data distribution to SCells.
Resource-based adaptive downlink scheduling	RES_BASED_DL_ADAPT_SCH_SW option of the NRDUCellDISch.DISchAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Enabling resource-based adaptive scheduling for massive MIMO increases the DU delay and has other impacts. These impacts may increase if resource-based adaptive downlink scheduling is also enabled. When both experience-based adaptive scheduling for massive MIMO and resource-based adaptive downlink scheduling are enabled, the first packet delay (indicated by the N.Traffic.DL.RlcFirstPktDelay.Time counter) may increase.
CA Performance Boost Based on Multi-dimensional Optimization	CA_PRFM_BOOST_BSD_ON_MD_OPT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	If resource-based adaptive scheduling for massive MIMO is enabled for cells and then CA Performance Boost Based on Multi-dimensional Optimization is also enabled for these cells, the benefits of CA Performance Boost Based on Multi-dimensional Optimization may decrease.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- Adaptive CSI-RS multi-beam adjustment
3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- Adaptive multi-stream perception
3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units can be used. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable low-frequency RF modules with 32T32R and above can be used. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Cells

AHR Introduction requires that the **NRDUCellTrp.TrpType** parameter be set to **MASTER_DM_MIMO** or **MASTER_DM_MIMO_LIGHT** when the **NRDUCell.LampSiteCellFlag** parameter is set to **YES**.

Adaptive CSI-RS multi-beam adjustment has the following requirements:

- The **NRDUCellCsirs.CsirsBeamType** parameter must be set to **TYPE0** or **TYPE1**.
- The **NRDUCellCsirs.FdCsiPeriod** parameter cannot be set to **SLOT40** for any 32T32R cell and to **SLOT40** or **SLOT80** for any non-32T32R cell.
- The **NRDUCellTrpBeam.ConnModeCoverageScenario** parameter cannot be set to **SCENARIO_1**.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-4 describes the parameters used for function activation.

Table 4-4 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
AHR Switch	NRDUCellFeatureSw. <i>AhrSwitch</i>	Select the AHR_PHASE1_SW option when functions related to AHR Introduction need to be enabled.
CSI-RS Beam Type	NRDUCellCsirs.<i>CsirsBeamType</i>	Use the recommended value.
CSIRS Cell Resource Number	NRDUCellCsirs.<i>CsirsCellResourceNum</i>	Set this parameter to FD_RESOURCE when adaptive CSI-RS multi-beam adjustment needs to be enabled.
FD CSI Period	NRDUCellCsirs.<i>FdCsirPeriod</i>	Use the recommended value.
Flexible and Dynamic CSI-RS Active UE Thld	NRDUCellCsirs.<i>FdCsirsActiveUeThld</i>	The value 60 is recommended when adaptive CSI-RS multi-beam adjustment needs to be enabled.
Flexible and Dynamic CSI-RS Path Loss Thld	NRDUCellCsirs.<i>FdCsirsPathLossThld</i>	The value 140 is recommended when adaptive CSI-RS multi-beam adjustment needs to be enabled.
CSI Switch	NRDUCellCsirs.<i>CsiSwitch</i>	When adaptive CSI-RS multi-beam adjustment needs to be enabled, it is recommended that the CSIRS_INTRF_STATIC_AVOID_SW option be selected for surrounding cells for which adaptive CSI-RS multi-beam adjustment is not enabled.
Downlink Initial MCS Adjust Value	NRDuCellPdsch.<i>DlInitialMcsAdjValue</i>	The value 0 is recommended when adaptive CSI-RS multi-beam adjustment needs to be enabled.
Fixed AMC Step Value	NRDUCellPdsch.<i>FixedAmcStepValue</i>	The value 40 is recommended when adaptive CSI-RS multi-beam adjustment needs to be enabled.

Parameter Name	Parameter ID	Setting Notes
Service Experience Algorithm Switch	NRDUCellServExp.ServiceExpAlgoSwitch	Select the EXP_BASED_MM_ADAPT_SCH_SW and RES_BASED_MM_ADAPT_SCH_SW options when adaptive multi-stream perception needs to be enabled.
DL Delay Schedule Buffer Threshold	NRDUCellPdsch.DlDelaySchBufferThld	Use the recommended value.
Downlink Schedule Opt Time Threshold	NRDUCellPdsch.DlSchOptTimeThld	Use the recommended value if adaptive multi-stream perception needs to be enabled.
DL Adaptive Scheduling Time Threshold	NRDUCellDISch.DlAdaptSchTimeThld	The value 30 is recommended when resource-based adaptive scheduling for massive MIMO is required in NSA and SA hybrid networking. Use the default value in other scenarios.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

To reduce interference to cells for which AHR Introduction is enabled, static avoidance of CSI-RS interference can be enabled for surrounding cells for which AHR Introduction is not enabled.

```
//Enabling static avoidance of CSI-RS interference
MOD NRDUCELLCSIRS: NrDuCellId=1, CsiSwitch=CSIRS_INTRF_STATIC_AVOID_SW-1;
```

Activation Command Examples

```
//Enabling AHR Introduction
MOD NRDUCELLFEATURESW: NrDuCellId=0, AhrSwitch=AHR_PHASE1_SW-1;
//Optional Setting the CsirsCellResourceNum, CsirsBeamType, FdCsiPeriod, FdCsirsActiveUeThld, and
FdCsirsPathLossThld parameters when adaptive CSI-RS multi-beam adjustment is required
MOD NRDUCELLCSIRS: NrDuCellId=0, CsirsCellResourceNum=FD_RESOURCE, CsirsBeamType=TYPE0,
FdCsiPeriod=SLOT160, FdCsirsActiveUeThld=60, FdCsirsPathLossThld=140;
//Setting parameters related to downlink MCS adjustment
MOD NRDUCELLPDSCH: NrDuCellId=0, FixedAmcStepValue=40, DlInitialMcsAdjValue=0;
```

```
//(Optional) Selecting the EXP_BASED_MM_ADAPT_SCH_SW option and setting the DIDelaySchBufferThld parameter when adaptive multi-stream perception is required  
MOD NRDUCELLSERVEXP: NrDuCellId=0, ServiceExpAlgoSwitch=EXP_BASED_MM_ADAPT_SCH_SW-1;  
MOD NRDUCELLPDSCH: NrDuCellId=0, DIDelaySchBufferThld=1;  
//(Optional) Selecting the RES_BASED_MM_ADAPT_SCH_SW option and setting the DISchOptTimeThld parameter when adaptive multi-stream perception is required  
MOD NRDUCELLSERVEXP: NrDuCellId=0, ServiceExpAlgoSwitch=RES_BASED_MM_ADAPT_SCH_SW-1;  
MOD NRDUCELLPDSCH: NrDuCellId=0, DISchOptTimeThld=300;  
//(Optional) Setting parameters when resource-based adaptive scheduling for massive MIMO is used in NSA and SA hybrid networking  
MOD NRDUCELLDLSCH: NrDuCellId=0, DIAaptSchTimeThld=30;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Using the default settings for CSI-RS beams  
MOD NRDUCELLCSIRS: NrDuCellId=0, CsirsCellResourceNum=4_RESOURCE;  
//Disabling AHR Introduction  
MOD NRDUCELLFEATURESW: NrDuCellId=0, AhrSwitch=AHR_PHASE1_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Adaptive CSI-RS multi-beam adjustment has taken effect if the value of the *periodicityAndOffset* IE in the RRC reconfiguration message is the same as the value of the **NRDUCellCsirs.FdCsiPeriod** parameter after this function is enabled.

If the PRB usage is greater than 60% but less than or equal to 95% for an extended period of time, observe the average downlink UE throughput before and after the resource-based adaptive scheduling for massive MIMO function is enabled. If the average downlink UE throughput increases, this function has taken effect.

4.4.3 Network Monitoring

The following indicators can be used to observe the average downlink UE throughput and average downlink cell throughput, respectively:

- **User Downlink Average Throughput (DU)**
- **Cell Downlink Average Throughput (DU)**

The network monitoring of this feature also involves other indicators whose definitions are described in [Table 4-5](#).

Table 4-5 Definitions of indicators related to AHR Introduction

Indicator	Definition
Average CQI	$\frac{\text{SUM}(k \times \text{N.ChMeas.CQI.SingleCW}.k)}{\text{SUM}(\text{N.ChMeas.CQI.SingleCW}.k)}$ $k = 0 \text{ to } 15$
Proportion of CQIs with a value of 0	$\frac{\text{N.ChMeas.CQI.SingleCW}.0}{\text{SUM}(\text{N.ChMeas.CQI.SingleCW}.k)}$ $k = 0 \text{ to } 15$
Average downlink MCS index	$\frac{\text{SUM}(k \times \text{N.ChMeas.PDSCH.MCS}.k)}{\text{SUM}(\text{N.ChMeas.PDSCH.MCS}.k)}$ $k = 0 \text{ to } 28$
Downlink PRB usage	$\text{N.PRB.DL.Used.Avg} / \text{N.PRB.DL.Avail.Avg}$
SgNB Mod Required Succ Rate	$\frac{\text{N.NsaDc.SgNB.Mod.Required.Succ}}{\text{N.NsaDc.SgNB.Mod.Required.Att}}$
DU delay (ms)	$\frac{(\text{N.QoS.DL.PktDelayAirInterface.Time} / \text{N.QoS.DL.PktDelayAirInterface.Num} + \text{N.QoS.DL.PktDelaygNBDU.Time} / \text{N.QoS.DL.PktDelaygNBDU.Num})}{1000}$

5 AHR Experience Turbo 2.0

5.1 Principles

AHR Experience Turbo 2.0 provides the following functions:

- **SRS interference coordination** reduces SRS interference and improves SRS quality and weight accuracy, thereby increasing the downlink user-perceived rate.
- **Precise AMC** improves MCS accuracy and optimizes MIMO pairing performance, thereby improving spectral efficiency and increasing the downlink user-perceived rate.

This feature is controlled by the **AHR_EXP_TURBO_PHASE2_SW** option of the **NRDUCellFeatureSw.AhrSwitch** parameter.

5.1.1 SRS Interference Coordination

In massive MIMO scenarios, the gNodeB can estimate comprehensive uplink channel characteristics by using SRSs. On TDD networks, the gNodeB can even deduce downlink channel characteristics from uplink channel characteristics since uplink and downlink channels are reciprocal. Massive MIMO channel estimation is affected by the accuracy and effectiveness of SRS measurement, which in turn is strongly affected by SRS interference. Against this backdrop, SRS interference coordination is introduced.

SRS interference coordination uses multiple antennas to identify and detect strong and burst SRS interference from neighboring cells. As shown in [Figure 5-1](#), after identifying and detecting the interference, the gNodeB uses technologies such as interference cancellation (IC), interference coordination, power control, and interference suppression to reduce SRS interference and improve SRS quality and weight accuracy, thereby increasing the downlink user-perceived rate.

Figure 5-1 SRS interference coordination

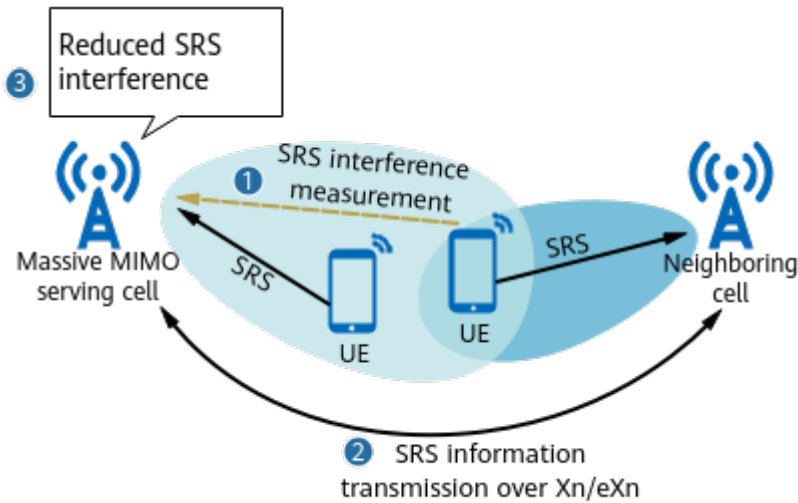


Table 5-1 describes the sub-functions of SRS interference coordination.

Table 5-1 Sub-functions of SRS interference coordination

Function Name	Function Description	Function Switch	Description
SRS IC	This function is used to demodulate interfering SRSs from neighboring cells and eliminate them from uplink signals received in the massive MIMO serving cell. This reduces or even eliminates the SRS interference from neighboring cells.	SRS_IC_SW option of the NRDUCellSrs.SrsDetectionA lgoSwitch parameter	After SRS IC is enabled, it takes effect only when the SRS interference in a cell is greater than the threshold specified by the NRDUCellSrsMeas.SrsIntrfThld parameter. If the physical cell identifier (PCI) mod 30 values of the serving cell and a neighboring cell are the same, the serving cell cannot distinguish SRSs intended for the neighboring cell. As a result, SRS interference from the neighboring cell cannot be eliminated.

Function Name	Function Description	Function Switch	Description
SRS joint power control	This function selects interfering neighboring cells and UEs by performing neighboring cell measurement, calculates a new target signal to interference plus noise ratio (SINR) for SRS power control based on the interference strength, and then adjusts the SRS transmit power of interfering UEs in neighboring cells according to the new target SINR.	SRS_JOINT_PC_SW option of the NRDUCellULPcConfig.UlPwrCtrlAlgoSwitch parameter	If the PCI mod 30 values of the serving cell and a neighboring cell are the same, the serving cell cannot distinguish SRSs intended for the neighboring cell. As a result, SRS joint power control does not take effect.
SRS beam-specific noise reduction	This function reduces SRS interference at the beam level, instead of at the UE level, to improve channel quality and weights for UEs.	BEAM_SPECIFIC_SRS_DENOISE_SW option of the NRDUCellSrs.SrsDetectionAlgoSwitch parameter	None
SRS noise reduction for low-SNR channels	This function performs denoising for UEs at a medium or long distance from the cell center to reduce burst interference to SRSs and improve channel quality and weights for these UEs.	LOW_SNR_CHANNEL_DENOISE_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeEnhanceSwitch parameter	None

5.1.2 Precise AMC

Precise AMC optimizes link performance and time-frequency resources for massive MIMO cells. It improves the MCS accuracy, optimizes MIMO pairing performance, and improves the spectral efficiency and downlink user-perceived rate.

The sub-functions described in [Table 5-2](#) select the optimal MCS for UEs and optimize pairing performance to achieve link adaptation and optimal spectral efficiency.

Table 5-2 Sub-functions for achieving link adaptation and optimal spectral efficiency

Function Name	Function Description	Function Switch
Downlink tail packet scheduling optimization	Background: Excessive MCS index reduction for rate matching or small-packet services may cause tail packet fragmentation. If tail packet fragmentation occurs, more PRB resources will be occupied, affecting the downlink user-perceived rate. (A tail packet refers to the last data packet before the buffer is empty.) Principles: The scheduler appropriately adjusts MCS indexes for UEs for which MCS index reduction is enabled for rate matching or small-packet services, preventing tail packet fragmentation.	TAIL_PKT_SCH_OPT_SW option of the NRDUCellPdsch.DlPdschAlgoSwitch parameter
Downlink MCS index adjustment optimization	Background: The MCS adjustment value in downlink adaptation is adjusted based on the hybrid automatic repeat request (HARQ) feedback. However, reducing the MCS index for small packets increases the proportion of ACKs in the HARQ feedback. As a result, the MCS index adjusted based on the HARQ feedback does not accurately reflect the current channel quality. Principles: When MCS index reduction is triggered for UEs running small-packet services, the MCS adjustment step is adaptively changed based on initial block error rate (IBLER) convergence to obtain precise MCS indexes for UEs.	DL_MCS_ADJ_OPT_SW option of the NRDUCellDlAmc.DlAmcAlgoSw parameter
Rank and SINR estimation optimization	Background: MCS indexes for UEs extremely close to the cell center may exceed the maximum value in the MCS index table. This can cause inaccurate rank and MCS selection due to an excessively high SINR. Principles: When the gNodeB estimates ranks and SINRs using SRS-based weights and the MCS indexes for some UEs exceed the maximum value in the MCS index table, the gNodeB performs compensation for such MCS indexes. In addition, the gNodeB considers the expected block error rate (BLER) difference between different ranks when estimating the spectral efficiency of each rank. This ensures more accurate rank and MCS selection for UEs.	RANK_AND_SINR_ESTIMATE_OPT_SW option of the NRDUCellDlRank.DlRankSelAlgoSw parameter

Function Name	Function Description	Function Switch
Intelligent MCS optimization for downlink SU-MIMO	Background: Due to interference and traffic fluctuations on the live network, MCS selection does not match the actual channel conditions. Principles: This function enables the gNodeB to start model training to select the optimal MCS based on UE characteristics, thereby increasing the average downlink UE throughput in the cell. The status of the trained model can be monitored using the prediction error. (The upper limit of the prediction error is specified by the NRDUCellAiAlgo.DISuPredictErrorUpperLimit parameter).	DL_SU_MCS_INTE L_OPT_SW option of the NRDUCellAiAlgo.AiAmcAlgoSwitch parameter

The sub-functions described in [Table 5-3](#) select optimal time-frequency resources for UEs to minimize time-frequency resource consumption.

Table 5-3 Sub-functions for minimizing time-frequency resource consumption

Function Name	Function Description	Function Switch
Precise RB control for downlink retransmissions	Background: By using HARQ during downlink scheduling for retransmissions, a received error data packet is stored in a HARQ buffer and combined with the subsequently received retransmitted data packet for more reliable decoding. However, if both initial transmissions and retransmissions use the same PRB resources, resource wastage occurs. Principles: To improve spectral efficiency, the retransmission bit rate can be increased during downlink scheduling for retransmissions. This reduces the number of PRBs used for retransmission scheduling, freeing them up for initial transmissions of other UEs.	DL_RETRANS_PRECISE_RB_CTRL_SW option of the NRDUCellDISch.DISchAlgoSwitch parameter

Function Name	Function Description	Function Switch
Downlink radio link control (RLC) status report awareness scheduling	Background: When the acknowledged mode (AM) is used for RLC transmission, the RLC status reports of uplink data need to be scheduled in the downlink. The scheduling of these reports is performed separately, which consumes CCE resources. This will affect downlink user experience when PDCCH CCE resources are insufficient or the downlink PRB usage is relatively high. Principles: When PDCCH CCE resources are insufficient or the downlink PRB usage is relatively high, RLC ACK status reports can be transmitted with downlink packets provided that these reports were originally to be transmitted separately and the wait timer does not expire.	DL_RLC_STATUS_RPT_AWARE_SCH_SW option of the NRDUCellDisch.DISchAlgoSwitch parameter
Downlink slot awareness scheduling	Background: In TDD, some symbols in self-contained slots are used for uplink and guard period (GP), which reduces the number of symbols available for PDSCH scheduling and the amount of data that can be scheduled. Principles: Scheduling of UEs with a relatively large buffer size can be postponed to non-self-contained slots when the downlink PRB usage is less than 80% or PDCCH CCE resources are sufficient (the PDCCH CCE allocation failure rate is less than 30%). For details about self-contained slots, see <i>Standards Compliance</i>.	DL_SLOT_AWARE_SCH_SW option of the NRDUCellDisch.DISchAlgoSwitch parameter

5.2 Network Analysis

5.2.1 Benefits

AHR Experience Turbo 2.0 increases the average downlink UE throughput when the downlink load is medium or high (for example, when the downlink PRB usage is greater than 30%) and SRS interference from neighboring cells exists (for example, when the value of **N.SRS.NI.Avg** is greater than -110 dBm). The actual benefit depends on factors such as interference and network load. The benefit is proportional to the amount of SRS interference from neighboring cells, the proportion of UEs in overlapping areas, the IBLER, and the downlink PRB usage, and inversely proportional to the proportion of tail packets.

In scenarios where the load is either light (for example, with a PRB usage less than 30%) or heavy (for example, with a PRB usage greater than 70%), some sub-functions take effect less often and provide lower or even no benefits.

For the definitions of related indicators, see [5.4.3 Network Monitoring](#).

In terms of the benefits of SRS IC and SRS joint power control:

- SRS IC:
 - If UEs in a cell use different SRS bandwidths, the benefits of SRS IC decrease. The more similar the proportions of UEs with different SRS bandwidths are, the larger the decrease. In such cases, this function may not provide benefits; however, it will not cause negative impacts.
 - It is recommended to enable this function in intra-frequency massive MIMO cells providing contiguous coverage. This function delivers no benefits in other scenarios.
- SRS joint power control: To prevent negative impacts from occurring, it is recommended to enable this function only in intra-frequency massive MIMO cells providing contiguous coverage.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

Table 5-4 describes the network impacts of SRS interference coordination.

Table 5-4 Network impacts of SRS interference coordination

Item	Impact
Uplink spectral efficiency	Due to improved downlink performance, the number of uplink TCP response packets may increase. If this occurs, the uplink spectral efficiency and average uplink UE throughput will decrease.
Uplink throughput	
Downlink throughput	• When tail packets account for an excessively large proportion of data volume in light- or medium-load scenarios, the downlink cell throughput may increase due to the improved SRS channel quality. However, the downlink UE throughput may decrease due to the increased proportion of tail packet data volume. • For lightly loaded cells in contiguous coverage areas, the SRS SINR may be negatively affected. If this occurs, the downlink cell throughput and downlink UE throughput will decrease.
SRS SINR	
SRB PRB usage	The number of RRC reconfiguration messages may increase. If this occurs, the SRB PRB usage will increase.

The following lists the network impacts of precise AMC:

- During model learning, the values of the following indicators may fluctuate within a short period of time:
 - Average downlink cell throughput
 - Average downlink MCS index in a cell
 - Downlink PRB usage in a cell
 - Average downlink user-perceived rate
- Some other network impacts also exist, as described in [Table 5-5](#).

Table 5-5 Other network impacts of precise AMC

Item	Impact
CPU usage	The CPU usage (VS.BBUBoard.CPULoad.Mean) may increase.
In the downlink: rank, MCS, BLER, first packet delay at the RLC layer, PRB usage, number of PRBs for MU pairing, and throughput	• Precise AMC affects UE downlink selection. It may increase the proportion of downlink rank 3 and rank 4, downlink IBLER and RBLER, first packet delay at the RLC layer, and the number of samples in the low-throughput range and may decrease the average downlink MCS index, downlink PRB usage, and number of PRBs for downlink MU pairing in a cell. • Precise AMC does not improve the average downlink UE throughput when most UEs are located at the cell center or at a medium distance from the cell center (for example, when the average CQI in the cell is greater than 12) and UEs use a single traffic model (for example, when the proportion of tail packet transmission duration is less than 20% or greater than 85%). • Due to improved downlink performance, the number of uplink TCP response packets may increase. If this occurs, the CCE aggregation level and the PDCCH CCE usage will increase, and the average uplink throughput will decrease. If CCE resources are insufficient at the same time, the average downlink throughput will decrease.
In the uplink: throughput	
Service drop rate	The service drop rate of a cell may slightly increase when the downlink IBLER increases.

For the definitions of related indicators, see [5.4.3 Network Monitoring](#).

5.3 Requirements

5.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-061201	AHR Experience Turbo 2.0	NR0S00AET200	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
AHR Introduction	AHR_PHASE1_SW option of the NRDUCellFeatureSw.AhrSwitch parameter	<i>Massive MIMO AHR (Low-Frequency TDD)</i>	None
UE-characteristic-based SRS period adaptation	USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter	<i>Channel Management</i>	SRS IC, SRS joint power control, SRS beam-specific noise reduction, and SRS noise reduction for low-SNR channels require UE-characteristic-based SRS period adaptation to be enabled.
Intra-frequency neighboring cell measurement	INTRA_FREQ_NC_ELL_STAT_SW option of the NRDUCellCollabServ.NeighborNrCellMgmtSw parameter	None	SRS IC and SRS joint power control require intra-frequency neighboring cell measurement to be enabled.

Function Name	Function Switch	Reference	Description
PDSCH MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	SRS beam-specific noise reduction is dependent on PDSCH MU-MIMO.
Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	For LampSite base stations, distributed massive MIMO must be enabled.
SRS Po adaptation	PO_SRS_ADAPT_SW option of the NRDUCellULPcConfig.ULPwrCtrlAlgoSwitch parameter	<i>Power Control</i>	SRS joint power control requires SRS Po adaptation to be enabled.

5.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
SRS Wideband Index Csr	NRDUCellSrs.SrsWideBandIndexCsr	None	After SRS IC is enabled, this parameter must be set to a value greater than 21 .
High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	High-speed Railway Superior Experience cannot be enabled together with the following functions: SRS IC, SRS joint power control, and intelligent MCS optimization for downlink SU-MIMO.
Hyper Cell	NRDUCell.NrDualCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	Neither SRS interference coordination nor SRS beam-specific noise reduction can be enabled for hyper cells.

Function Name	Function Switch	Reference	Description
Cell Combination	NRDUCell.NrDualCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	Neither SRS interference coordination nor SRS beam-specific noise reduction can be enabled for combined cells.
SUL cell	NRDUCell.DuplexMode set to CELL_SUL	None	None

5.3.2.3 Function Impacts

Function Name	Function Switch	Reference	Description
SRS filtering-based downlink beamforming weight optimization	DL_BF_WEIGHTFILTER_OPT_SW option of the NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch parameter	<i>FWA</i>	When both SRS noise reduction for low-SNR channels and SRS filtering-based downlink beamforming weight optimization are enabled, only the former takes effect.
High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	SRS noise reduction for low-SNR channels does not take effect for high-speed cells.
Downlink 8-layer transmission for a single FWA user	DL_SU_EIGHT_LAYER_SW option of the NRDUCellDlRank.DlRankSelAlgoSw parameter	<i>FWA</i>	If downlink 8-layer transmission for a single FWA user has taken effect, intelligent MCS optimization for downlink SU-MIMO does not take effect for the user.
Downlink 8-layer transmission for a single FWA user	DL_SU_EIGHT_LAYER_SW option of the NRDUCellDlRank.DlRankSelAlgoSw parameter	<i>FWA</i>	If downlink 8-layer transmission for a single FWA user has taken effect, SRS beam-specific noise reduction does not take effect for the user.

Function Name	Function Switch	Reference	Description
DL 1024QAM	DL_1024QAM_SW option of the NRDUCellDLAmc.DlModulationSchemeSw parameter	<i>Modulation Schemes</i>	Intelligent MCS optimization for downlink SU-MIMO does not take effect for UEs for which the DL 1024QAM MCS index table is selected.
Power optimization for downlink retransmissions	DL_RETRANS_POWER_OPT_SW option of the NRDUCellChnPower.ChnPwrAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	Precise RB control for downlink retransmissions preferentially takes effect when it is enabled together with power optimization for downlink retransmissions. Power optimization for downlink retransmissions can take effect only when precise RB control for downlink retransmissions is not in effect.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

Table 5-6 describes the base station model requirements of SRS interference coordination and precise AMC.

Table 5-6 Base station model requirements of SRS interference coordination and precise AMC

Function	Sub-Function	Base Station Model
SRS interference coordination	SRS beam-specific noise reduction and SRS noise reduction for low-SNR channels	3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Function	Sub-Function	Base Station Model
	Others	3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910. - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.
Precise AMC	Intelligent MCS optimization for downlink SU-MIMO	3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
	Others	3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910. - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units can be used. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

UMPTe boards support intelligent MCS optimization for downlink SU-MIMO only when they work in a single mode. Other NR TDD-capable main control boards support this function when they work in a single mode or multiple modes.

RF Modules

- For SRS IC in SRS interference coordination:
All NR TDD-capable eCPRI-compliant low-frequency RF modules with 32T32R and above can be used. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.
- For other functions:
All NR TDD-capable low-frequency RF modules with 32T32R and above can be used. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Cells

Table 5-7 describes the cell requirements of SRS interference coordination and precise AMC.

Table 5-7 Cell requirements of SRS interference coordination and precise AMC

Function	Sub-Function	Cell Requirement
SRS interference coordination	SRS IC	- The NRDUCell.NrDuCellNetworkingMode parameter is set to NORMAL_CELL or FUSION_CELL. - The NRDUCell.DlBandwidth parameter is set to CELL_BW_40M, CELL_BW_50M, CELL_BW_60M, CELL_BW_70M, CELL_BW_80M, CELL_BW_90M, or CELL_BW_100M. - When the NRDUCell.LampSiteCellFlag parameter is set to YES, the NRDUCellTrp.TrpType parameter is set to MASTER_DM_MIMO.
	SRS joint power control	- The NRDUCell.DlBandwidth parameter is set to CELL_BW_80M or CELL_BW_100M. - The NRDUCell.DeployPosition parameter is set to DEPLOY_INTEGRATED. - When the NRDUCell.LampSiteCellFlag parameter is set to YES, the NRDUCellTrp.TrpType parameter is set to MASTER_DM_MIMO or MASTER_DM_MIMO_LIGHT. - SRS joint power control requires the following conditions to be met to ensure cell performance: - The PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter is selected. - The RLC_TIMER_ADAPT_WITH_SR_PRD_SW option of the NRDUCellServExp.RlcSwitch parameter is selected. - The RLC_PARAM_ADAPT_SW option of the NRDUCellServExp.RlcSwitch parameter is selected. - The AGG_LVL_INIT_SELECT_OPT_SW option of the NRDUCellPdcch.PdcchAlgoEnhSwitch parameter is selected. - The NRDUCellPdcch.PdcchAggLvlAdaptPol parameter is set to SEPARATE_ADAPT. - The SR_SMART_SCH_SW option of the NRDUCellPusch.UlPuschAlgoSwitch parameter is selected. - The NRDUCellPusch.RetransBsrTimer parameter is set to SF320. - The NRDUCell.NrDuCellNetworkingMode parameter is set to NORMAL_CELL or FUSION_CELL.

Function	Sub-Function	Cell Requirement
	SRS noise reduction for low-SNR channels	When the NRDUCell.LampSiteCellFlag parameter is set to YES , the NRDUCellTrp.TrpType parameter is set to MASTER_DM_MIMO .
	Others	None
Precise AMC	Rank and SINR estimation optimization	The NRDUCellPdcch.PdcchAggLvlAdaptPol parameter is set to SEPARATE_ADAPT .
	Intelligent MCS optimization for downlink SU-MIMO	- The NRDUCell.NrDuCellNetworkingMode parameter is set to NORMAL_CELL. - The NRDUCell.HighSpeedFlag parameter is set to LOW_SPEED. - The NRDUCellPdcch.PdcchAggLvlAdaptPol parameter is set to SEPARATE_ADAPT.
	Others	None

5.3.5 Networking

- For SRS IC and SRS joint power control in SRS interference coordination:
The Xn interface between gNodeBs must work properly.
- For other functions:
None

5.3.6 Others

None

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

[Table 5-8](#) describes the parameters used for function activation.

Table 5-8 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
AHR Switch	NRDUCellFeatureSw.AhrSwitch	AHR_EXP_TURBO_PHASE2_SW	Select this option when functions related to AHR Experience Turbo 2.0 need to be enabled.
SRS Detection Algorithm Switch	NRDUCellSrs.SrsDetectionAlgoSwitch	SRS_IC_SW	Select this option when SRS interference coordination needs to be enabled.
SRS Interference Threshold	NRDUCellSrsMeas.SrsInterfThld	None	Use the recommended value. You can set this parameter to a larger value if there is a high proportion of moving UEs in the cell where SRS interference coordination needs to be enabled.
SRS Detection Algorithm Switch	NRDUCellSrs.SrsDetectionAlgoSwitch	BEAM_SPECIFIC_SRS_DENOISE_SW	It is recommended that this option be selected when SRS interference coordination needs to be enabled.
Uplink Power Control Algorithm Switch	NRDUCellULPcConfig.UlPwrCtrlAlgoSwitch	SRS_JOINT_PC_SW	It is recommended that this option be selected when there is high SRS interference between heavy traffic cells. The default setting is recommended in other scenarios.

Parameter Name	Parameter ID	Option	Setting Notes
Intrf UE SRS Pwr Ctrl Min SINR Target	NRDUCellULPcConfig.IntrfUeSrsPcMinSinrTarget	None	The value 50 is recommended when SRS interference coordination is enabled. The parameter setting can be optimized based on the SRS SINR distribution on the network to obtain more benefits.
Max SRS PO Adjustment Amount	NRDUCellULPcConfig.MaxSrsPoAdjustAmount	None	It is recommended that this parameter be set to 6 when SRS interference coordination needs to be enabled. If this parameter is set to 10 , the service drop rate may increase because the SRS power adjustment step is too large.
Adaptive Edge Experience Enhance Switch	NRDUCellAlgoSwitch.AdaptiveEdgeExpEnhSwitch	LOW_SNR_CHN_DENOISE_SW	In FWA scenarios, it is recommended that this option be selected when SRS interference coordination needs to be enabled. In other scenarios, selecting this option may have negative impacts and therefore the default setting is recommended.
Downlink PDSCH Algorithm Switch	NRDUCellPdsch.DLPdschAlgoSwitch	TAIL_PKT_SCH_OPT_SW	Select this option when precise AMC needs to be enabled.

Parameter Name	Parameter ID	Option	Setting Notes
Downlink AMC Algorithm Switch	NRDUCellDlAmc.DlAmcAlgoSw	DL_MCS_ADJ_OPT_SW	It is recommended that all these options be selected when precise AMC needs to be enabled. If only the DL_MCS_ADJ_OPT_SW option is selected, the average downlink UE throughput may decrease.
DL Rank Selection Algorithm Switch	NRDUCellDlRank.DlRankSelAlgoSw	RANK_AND_SINR_ESTIMATE_OPT_SW	
AI AMC Algorithm Switch	NRDUCellAiAlgo.AiAmcAlgoSwitch	DL_SU_MCS_INTERL_OPT_SW	
DL SU-MIMO Predict Error Upper Limit	NRDUCellAiAlgo.DlSuPredictErrorUpperLimit	None	Use the recommended value.
Downlink Schedule Algorithm Switch	NRDUCellDlSch.DlSchAlgoSwitch	- DL_RLC_STATISTICS_RPT_AWARE_SCH_SW - DL_SLOT_AWARE_SCH_SW - DL_RETRANS_PRECISE_RB_CTRL_SW	It is recommended that all these options be selected when precise AMC needs to be enabled.

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling AHR Experience Turbo 2.0
MOD NRDUCELLFEATURESW: NrDuCellId=0, AhrSwitch=AHR_EXP_TURBO_PHASE2_SW-1;
//Enabling SRS IC and SRS beam-specific noise reduction
MOD NRDUCELLSRS: NrDuCellId=0,
SrsDetectionAlgoSwitch=SRS_IC_SW-1&BEAM_SPECIFIC_SRS_DENOISE_SW-1;
MOD NRDUCELLSRSMEAS: NrDuCellId=0, SrsIntrFThId=5;
```

```
//(Optional) Enabling SRS joint power control
MOD NRDUCELLULPCONFIG: NrDuCellId=0, UlpwrCtrlAlgoSwitch=SRS_JOINT_PC_SW-1;
MOD NRDUCELLULPCONFIG: NrDuCellId=0, IntrfUeSrsPcMinSinrTarget=50, MaxSrsPoAdjustAmount=6;
//(Optional) Using the default setting for SRS noise reduction for low-SNR channels
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, AdaptiveEdgeExpEnhSwitch=LOW_SNR_CHN_DENOISE_SW-0;
//Enabling downlink tail packet scheduling optimization
MOD NRDUCELLPDSCH: NrDuCellId=0, DlPdschAlgoSwitch=TAIL_PKT_SCH_OPT_SW-1;
//Enabling downlink MCS index adjustment optimization, downlink rank and SINR estimation optimization,
and intelligent MCS optimization for downlink SU-MIMO (Note: Intelligent MCS optimization for downlink
SU-MIMO is not supported by UMPTE boards working in multiple modes, and therefore it fails to be
enabled when such a board is used.)
MOD NRDUCELLDLAMC: NrDuCellId=0, DlAmcAlgoSw=DL_MCS_ADJ_OPT_SW-1;
MOD NRDUCELLDLRANK: NrDuCellId=0, DlRankSelAlgoSw=RANK_AND_SINR_ESTIMATE_OPT_SW-1;
MOD NRDUCELLAIALGO: NrDuCellId=0, AiAmcAlgoSwitch=DL_SU_MCS_INTEL_OPT_SW-1,
DlSuPredictErrorUpperLimit=5;
//Enabling downlink RLC status report awareness scheduling
MOD NRDUCELLDLSCH: NrDuCellId=0, DlSchAlgoSwitch=DL_RLC_STATUS_RPT_AWARE_SCH_SW-1;
//Enabling downlink slot awareness scheduling
MOD NRDUCELLDLSCH: NrDuCellId=0, DlSchAlgoSwitch=DL_SLOT_AWARE_SCH_SW-1;
//Enabling precise RB control for downlink retransmissions
MOD NRDUCELLDLSCH: NrDuCellId=0, DlSchAlgoSwitch=DL_RETRANS_PRECISE_RB_CTRL_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling AHR Experience Turbo 2.0
MOD NRDUCELLFEATURESW: NrDuCellId=0, AhrSwitch=AHR_EXP_TURBO_PHASE2_SW-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

If SRS interference (**N.SRS.NI.Avg**) is reduced after SRS interference coordination is enabled, this function has taken effect.

If the average PDSCH MCS index changes or the proportion of rank 3 and rank 4 increases, precise AMC has taken effect.

To check the status of the model corresponding to intelligent MCS optimization for downlink SU-MIMO, run the **DSP NRDUCELLAISCHMODEL** command with **AiCase** set to **DL_SU_AI_AMC**.

- The value "Unavailable" is displayed if the **DL_SU_MCS_INTEL_OPT_SW** option of the **NRDUCellAiAlgo.AiAmcAlgoSwitch** parameter is selected but the model cannot be launched due to reasons such as cell deactivation or license insufficiency.
- The value "Building" is displayed if the **DL_SU_MCS_INTEL_OPT_SW** option of the **NRDUCellAiAlgo.AiAmcAlgoSwitch** parameter is selected and the model starts to be built but has not been launched.
- The value "In Use" is displayed if the model has been launched and meets the preset accuracy requirements.

- The value "Updating" is displayed if the model has been suspended and a model update has been triggered because it did not meet the preset accuracy requirements.

5.4.3 Network Monitoring

The following indicators can be used to observe the average downlink UE throughput and average downlink cell throughput, respectively:

- **User Downlink Average Throughput (DU)**
- **Cell Downlink Average Throughput (DU)**

The network monitoring of this feature also involves other indicators whose definitions are described in [Table 5-9](#).

Table 5-9 Definitions of indicators related to AHR Experience Turbo 2.0

Indicator	Definition
Average downlink MCS index	$\frac{\text{SUM}(k \times \text{N.ChMeas.PDSCH.MCS}.k)}{\text{SUM}(\text{N.ChMeas.PDSCH.MCS}.k)}$ $k = 0 \text{ to } 28$
Downlink PRB usage	$\text{N.PR.B.DL.Used.Avg} / \text{N.PR.B.DL.Avail.Avg}$
Number of PRBs for downlink MU pairing	$\text{N.ChMeas.MIMO.DL.Pair.PR.B}$
SRB PRB usage	$(\text{N.PR.B.DL.Used.Avg} - \text{N.PR.B.DL.DrbUsed.Avg}) / \text{N.PR.B.DL.Avail.Avg}$
Average CQI	$\frac{\text{SUM}(k \times \text{N.ChMeas.CQI.SingleCW}.k)}{\text{SUM}(\text{N.ChMeas.CQI.SingleCW}.k)}$ $k = 0 \text{ to } 15$
Downlink IBLER	$(\text{N.DL.SCH.QPSK.ErrTB.Ibler} + \text{N.DL.SCH.16QAM.ErrTB.Ibler} + \text{N.DL.SCH.64QAM.ErrTB.Ibler} + \text{N.DL.SCH.256QAM.ErrTB.Ibler}) / (\text{N.DL.SCH.QPSK.TB} + \text{N.DL.SCH.16QAM.TB} + \text{N.DL.SCH.64QAM.TB} + \text{N.DL.SCH.256QAM.TB})$
First packet delay at the RLC layer	$\text{N.Traffic.DL.RlcFirstPktDelay.Time} / \text{N.Qos.DL.RlcFirstPktDelay.Num}$
Proportion of downlink tail packet transmission duration	$(\text{N.ThpTime.DL} - \text{N.ThpTime.DL.RmvLastSlot}) / \text{N.ThpTime.DL}$
Service drop rate of a cell	$\text{N.UECntx.AbnormRel} / (\text{N.UECntx.NormRel} + \text{N.UECntx.AbnormRel})$

6 AHR Capacity Upgrade 2.0

6.1 Principles

AHR Capacity Upgrade 2.0 provides the following functions:

- **High-resolution MU pairing** allows more suitable UEs to participate in MU pairing and increases the numbers of paired UEs and PRBs. This improves cell capacity and user experience.
- **Adaptive MU resource management** increases MU-MIMO resource usage and optimizes resource allocation. This enhances the overall MU pairing performance and further improves cell capacity and user experience.

This feature is controlled by the **AHR_CAPC_UPGRADE_PHASE2_SW** option of the **NRDUCellFeatureSw.AhrSwitch** parameter.

6.1.1 High-Resolution MU Pairing

In medium- and heavy-load scenarios, user experience is unsatisfactory due to insufficient cell resources. Given the same traffic volume, the benefits of MU-MIMO are proportional to the numbers of paired RBs and layers. High-resolution MU pairing is thus introduced to schedule UEs in a timely manner and improve user experience.

In TDD massive MIMO cells, high-resolution MU pairing optimizes PDSCH MU-MIMO and PDCCH MU-MIMO from multiple dimensions, such as UE grouping and the number of layers, so that more suitable UEs can participate in MU pairing, as shown in [Figure 6-1](#). This increases the numbers of paired UEs and PRBs, thereby improving the cell capacity and user experience.

Figure 6-1 Principles of high-resolution MU pairing

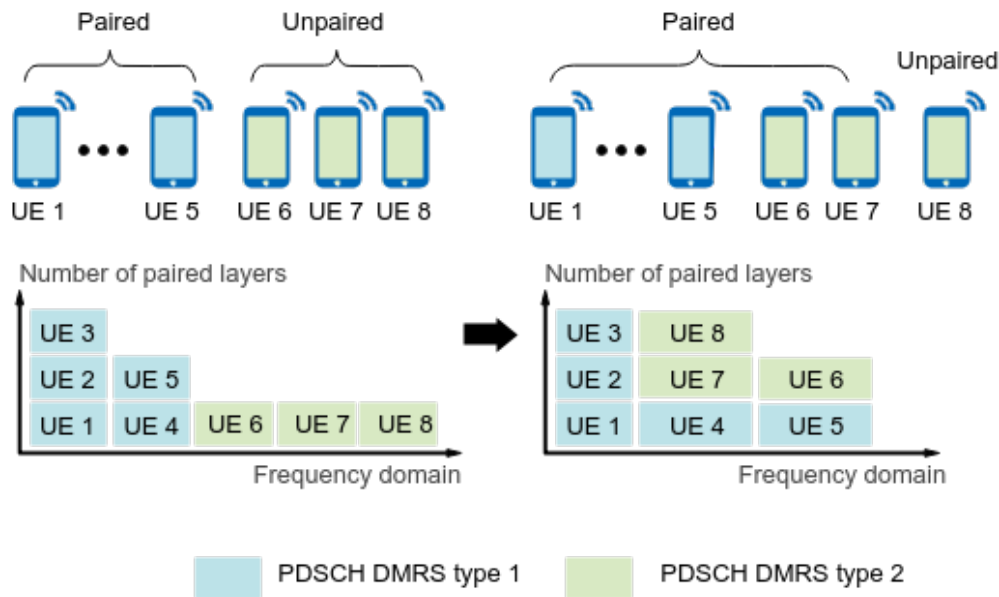


Table 6-1 describes the sub-functions of high-resolution MU pairing.

Table 6-1 Sub-functions of high-resolution MU pairing

Function Name	Function Description	Function Switch
Precise grouping optimization	This function optimizes the grouping of UEs to be paired to increase the number of UEs in the groups in a cell.	DL_PRECISE_GROUP_OPT_SW option of the NRDUCellIDIMimo.DLMuMimoSchSupplementSw parameter
MU-MIMO UE selection optimization	This function enables the gNodeB to filter UEs based on their data volumes and beam characteristics so that more suitable UEs can be paired for MU-MIMO.	MU_MIMO_USER_SELECT_OPT_SW option of the NRDUCellIDIMimo.DLMuMimoSchSupplementSw parameter
Downlink moving UE pairing optimization	This function optimizes pairing for moving UEs to decrease the IBLER and increase the MCS index of paired UEs, and reduce pairing losses for moving UEs. However, the number of paired layers may decrease.	DL_MOBILE_USER_PAIR_SW option of the NRDUCellIDIMimo.DLMuMimoSchSupplementSw parameter

Function Name	Function Description	Function Switch
MU pairing enhancement	This function allows the use of more SRS information for more accurate UE pairing, increasing the number of downlink paired layers. It is recommended that the NRDUCellPdsch.DIMuGatherOptSw parameter be set to ON so that more large-packet UEs can be paired. This function also involves certain thresholds specified by the following parameters: • NRDUCellPdsch.DLSrsMuMimoFreqSelThld: indicates the delay spread threshold for UEs to enter the pairing procedure. UEs whose delay spread is less than or equal to this threshold are more likely to enter the pairing procedure. • NRDUCellPdsch.DLSrsMuMimoPreSnrThld: indicates the threshold of the SRS SINR before equalization (SRS PreSINR) for UEs to enter the pairing procedure. UEs whose SRS PreSINR is greater than or equal to this threshold are more likely to enter the pairing procedure.	NRDUCellAlgoSwitch.MuMimoOptSw with
MU-MIMO spectral efficiency optimization	This function realizes power allocation and demodulation reference signal (DMRS) allocation between paired UEs to increase the downlink MU-MIMO pairing probability and the number of paired RBs.	SPECTRAL_EFFICIENCY_OPT_SW option of the NRDUCellDLIMimo.DLMuMimoSchSupplementSw parameter
PDCCH multi-dimension joint scheduling	This function allocates CCE resources in a centralized manner during PDCCH resource allocation, and preferentially performs space-domain pairing.	PDCCH_MULTI_DIM_JOINT_SCH_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter

6.1.2 Adaptive MU Resource Management

In medium- and heavy-load scenarios, user experience is unsatisfactory due to insufficient cell resources. Given the same traffic volume, more PDSCH resources for paired UEs results in lower interference between UEs, more accurate rank

selection, faster scheduling, and better user experience. Adaptive MU resource management is thus introduced.

In TDD massive MIMO cells, adaptive MU resource management optimizes PDSCH MU-MIMO from multiple dimensions, such as the power, weight, scrambling, and priority of UEs to be paired, to improve MU-MIMO resource usage and optimize resource allocation. This improves the overall MU pairing performance, cell capacity, and user experience.

Table 6-2 describes the sub-functions of adaptive MU resource management.

Table 6-2 Sub-functions of adaptive MU resource management

Function Name	Function Description	Function Switch
RBG-level power normalization adaptation for downlink MU-MIMO	This function enables RBG-specific power normalization in every slot to reduce the interference between UEs.	DL_MU_RBG_PWR_NORM_ADAPT_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter
Interference rejection precoding for downlink MU-MIMO in NLOS scenarios	This function optimizes the weights for paired UEs to better mitigate interference between these UEs. The related threshold and policy are specified by the following parameters: NRDUCellPdsch.SrsPresinrThld: indicates the SRS PreSINR threshold. When the SRS quality of a UE is greater than or equal to this threshold, this function can take effect. NRDUCellPdschPre-code.DIMuIRPrecodePol: indicates the interference rejection precoding policy for MU-MIMO. This function requires a UE to use two or four SRS ports.	DL_MU_NLOS_IR_PRECODING_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter
DMRS pseudorandom optimization	This function determines the scrambling policy for downlink DMRSs, decreasing the DMRS symbol overhead and increasing available PDSCH transmission resources.	DL_DMRS_SCRAMBLE_POLICY_SW option of the NRDUCellDmrs.DmrsSwitch parameter

Function Name	Function Description	Function Switch
Load-based adaptive downlink scheduling	With this function, the gNodeB automatically checks the cell load. If the gNodeB determines that the cell load is higher than the heavy-load PRB usage threshold for downlink scheduling^a, it will increase the scheduling priorities for UEs at the cell center or at a medium distance from the cell center. In addition, the minimum expected downlink rate of UEs^b can be configured. When the downlink rate of a cell-edge UE (CEU) is lower than the minimum expected value, the scheduling priority of this UE will be increased. This prevents service drops of CEUs due to overly low scheduling priorities. a: The heavy-load PRB usage threshold for downlink scheduling is specified by the NRDUCellDISch.DISchHeavyLoadPrbThld parameter. b: The minimum expected downlink rate of UEs is specified by the NRDUCellDISch.MinExpectedDLUeRate parameter.	NRDUCellDISch.DISchPolicy set to ADAPTIVE
Enhanced MU rank	This function selects appropriate ranks for UEs based on the number of paired layers and performance differences between layers. The NRDUCellDIRank.MuMimoRankEnhEqv-SinrThld parameter can also be set on the gNodeB. It indicates the SINR threshold that decides whether a high rank is selected for UEs to take part in pairing.	NRDUCellDIRank.DISrsMuMimoRankMode set to ENHANCE_MU_RANK

6.2 Network Analysis

6.2.1 Benefits

AHR Capacity Upgrade 2.0 increases the average downlink cell throughput and average downlink UE throughput in heavy-load scenarios (for example, when the PRB usage is greater than 70%). The actual benefit depends on factors such as the traffic model and network load and is proportional to the downlink PRB usage, the number of PRBs for downlink MU pairing, and the number of paired layers.

For AHR Capacity Upgrade 2.0 to achieve its optimal benefits, [AHR Experience Turbo 2.0](#) must also be enabled. Otherwise, AHR Capacity Upgrade 2.0 provides fewer or even no benefits.

For the definitions of related indicators, see [6.4.3 Network Monitoring](#).

6.2.2 Impacts

The impacts between this function and other functions are described in [6.3.2.3 Function Impacts](#).

Table 6-3 describes the network impacts of high-resolution MU pairing and adaptive MU resource management.

Table 6-3 Network impacts of high-resolution MU pairing and adaptive MU resource management

Item	Impact
Downlink throughput	- After the downlink MU-MIMO pairing rate increases, the downlink UE throughput may decrease in either of the following situations: (1) the data volume of tail packets accounts for an excessively large proportion in medium-load scenarios; (2) the channel quality of UEs is good and UEs are distributed in a centralized manner. - If the PDCCH CCE allocation success rate increases, the number of times PDCCH resources are scheduled with a high aggregation level increases. This decreases the gains in the average downlink cell throughput and average downlink UE throughput.
Uplink throughput	- After the downlink MU-MIMO pairing rate increases, the number of scheduled UEs in the downlink and the consumption of CCE resources will increase. If CCE resources for the uplink are limited, the CCE allocation failure rate for uplink transmission will increase and the uplink cell throughput and user-perceived rate will decrease. - After the downlink MU-MIMO pairing rate increases, the number of uplink status reports will increase. If PUSCH PRB resources are limited, the uplink cell throughput and user-perceived rate will decrease.
PRB usage	The PRB utilization may decrease in medium- and heavy-load scenarios. It fluctuates normally in light-load scenarios.
Downlink BLER	After the MU-MIMO pairing rate increases, the interference between paired UEs will increase and the downlink BLER in the cell will increase.
Average MCS index and average rank	After the downlink MU-MIMO pairing rate increases, the average downlink MCS index and average rank will decrease.
Access success rate	When a CEU accesses or is handed over to an MU-MIMO cell, the access success rate of the UE will slightly decrease.

In downlink heavy-load scenarios (the downlink PRB usage of a cell is greater than 80% or the downlink CCE usage is greater than 30%), load-based adaptive downlink scheduling in adaptive MU resource management causes impacts that are detailed in [Table 6-4](#).

Table 6-4 Network impacts of load-based adaptive downlink scheduling in downlink heavy-load scenarios

Indicator	Impact
Downlink BLER	The downlink IBLER may increase.
Service drop rate	The service drop rate in the cell may increase.
First packet delay at the RLC layer	The first packet delay at the RLC layer increases.
Downlink throughput	The fairness of resource allocation for UEs is affected. As a result, the downlink RLC throughput distribution changes significantly. Specifically, the number of samples increases in the low and medium-high ranges but decreases in the low-medium range.
Uplink throughput	In uplink heavy-load scenarios (the uplink PRB usage of a cell is greater than 80% or the uplink CCE usage is greater than 30%), the average uplink cell throughput and average uplink user-perceived rate decrease.
Uplink BLER and average uplink MCS index	The average uplink MCS index decreases, and the uplink IBLER and uplink RBLER may increase.

For the definitions of related indicators, see [6.4.3 Network Monitoring](#).

6.3 Requirements

6.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-061202	AHR Capacity Upgrade 2.0	NR0S00ACT200	per Cell

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
PDSCH MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch . MuMimoSwitch parameter	<i>MIMO (TDD)</i>	None
PDCCH MU-MIMO	PDCCH_MU_SW option of the NRDUCellAlgoSwitch . MuMimoSwitch parameter	<i>MIMO (TDD)</i>	PDCCH multi-dimension joint scheduling is dependent on PDCCH MU-MIMO.
Distributed massive MIMO	DM_MIMO_SERVICE_SWITCH option of the NRDUCellAlgoSwitch . DmMimoSwitch parameter	<i>Distributed Antenna Solutions (Low-Frequency TDD)</i>	For LampSite base stations, distributed massive MIMO must be enabled.
UE-characteristic-based SRS period adaptation	USER_CHARACTER_SRS_ADAPT_SW option of the NRDUCellSrs.SrsAlgoSwitch parameter	<i>Channel Management</i>	Interference rejection precoding for downlink MU-MIMO in NLOS scenarios requires UE-characteristic-based SRS period adaptation to be enabled.

6.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
High-speed Railway Superior Experience	NRDUCell.HighSpeedFlag	<i>High Speed Mobility</i>	Interference rejection precoding for downlink MU-MIMO in NLOS scenarios cannot be enabled together with High-speed Railway Superior Experience.

Function Name	Function Switch	Reference	Description
Hyper Cell	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL	<i>Hyper Cell</i>	DMRS pseudorandom optimization and interference rejection precoding for downlink MU-MIMO in NLOS scenarios cannot be enabled for hyper cells.
Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	DMRS pseudorandom optimization and interference rejection precoding for downlink MU-MIMO in NLOS scenarios cannot be enabled for combined cells.
SUL cell	NRDUCell.DuplexMode set to CELL_SUL	None	AHR Capacity Upgrade 2.0 cannot be enabled for SUL cells.
U-TDOA-based positioning and TOA fingerprint-based positioning	UTDOA_SW option of the NRCellAlgoSwitch.LcsSwitch parameter	<i>Positioning</i>	MU pairing enhancement cannot be enabled together with U-TDOA-based positioning or TOA fingerprint-based positioning.
Downlink noise reduction in MU beam domain	DL_MU_BEAM_NOISE_REDUCTION_SW option of the NRDUCellAlgoSwitch.DmMimoSwitch parameter	None	Interference rejection precoding for downlink MU-MIMO in NLOS scenarios cannot be enabled together with downlink noise reduction in MU beam domain.
Light pRRU channel calibration	LIGHT_PRRU_CHN_CALIB_SW option of the NRDUCellAlgoSwitch.ChnCalibSwitch parameter	None	MU pairing enhancement cannot be enabled together with light pRRU channel calibration.
Intra-base-station multi-user Co-MM	INTRA_GNB_MU_COMM_SW option of the NRDUCellCollabServ.MultiCellMimoSwitch parameter	None	Intra-base-station multi-user Co-MM cannot be enabled when interference rejection precoding for downlink MU-MIMO in NLOS scenarios is enabled and the NRDUCellPdschPrecode.DIMuIRPrecodePol parameter is set to MU_IRPUSER , MU_ALLUSER , or MU_ALLUSER_POWER_ENH .
DL MU Intra-Base Station Interference Rejection Precoding Policy	NRDUCellPdschPrecode.DIMuIRPrecodePol	None	

6.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDCCH symbol number adaptation	UE_PDCCH_SYM_NUM_ADAPT_SW option of the NRDUCellPdcch.PdcchAlgoExtSwitch parameter	<i>Channel Management</i>	When PDCCH multi-dimension joint scheduling is enabled in scenarios where PDCCH symbol number adaptation takes effect, the number of PDCCH symbols may not increase in a timely manner because the PDCCH CCE usage decreases. As a result, the uplink CCE resources are limited, and the uplink cell throughput and uplink user-perceived rate decrease.
Low-frequency TDD	PDCCH PMI-based weighting	NRDUCellPdcchAlgo.PdcchPrecoderEnhPolicy set to ENH_PMI	None	When PDCCH multi-dimension joint scheduling takes effect and PDCCH PMI-based weighting is enabled, the value of the NRDUCellPdcchAlgo.PdcchPrecoderEnhPreSnrThld parameter must be less than or equal to -10 dB.
Low-frequency TDD	Simulated load	NRDUCellSimulateLoad.TestMode set to TDD_SIMULATED_LOAD_MODE	None	When PDCCH multi-dimension joint scheduling is enabled in scenarios where simulated load takes effect, the downlink PDCCH CCE allocation success rate may decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink experience-based scheduling	DL_EXP_BASED_SCH_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	When downlink experience-based scheduling is in effect, the number of available downlink RBs in a cell decreases. As a result, the downlink PRB usage may fall below the threshold (that determines whether a cell is heavily loaded) for load-based adaptive downlink scheduling to take effect. Consequently, there are fewer opportunities for load-based adaptive downlink scheduling to produce its intended benefits.
Low-frequency TDD	Interference-based subband robust weights	INTRF_BASED_ROBUST_WEIGHT_SW option of the NRDUCellInterfIdent.ChnMeasCtrlSw parameter	<i>Interference Avoidance</i>	When both interference-based subband robust weights and load-based adaptive downlink scheduling are enabled, only load-based adaptive downlink scheduling takes effect.
Low-frequency TDD	Downlink 8-layer transmission for a single FWA user	DL_SU_EIGHT_LAYER_SW option of the NRDUCellDLRank.DlRankSelAlgoSw parameter	<i>FWA</i>	If downlink 8-layer transmission for a single FWA user has taken effect, interference rejection precoding for downlink MU-MIMO in NLOS scenarios does not take effect for the user.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink scheduling based on load and experience	LOAD_BASED_DL_EXP_SWITCH option of the NRDUCellDLISch.DLSchAlgoSwitch parameter	<i>Massive MIMO iBeam (Low-Frequency TDD)</i>	When downlink scheduling based on load and experience is enabled but MU-MIMO spectral efficiency optimization is disabled, UEs may be less likely to be paired. If this occurs, the cell throughput decreases.
Low-frequency TDD	Downlink experience satisfaction guarantee	DL_EXP_SATIS_GUARANTEE option of the NRDUCellFWaQos.SchAlgoSwitch parameter	<i>FWA</i>	When downlink experience satisfaction guarantee is in effect, the number of available downlink RBs in a cell decreases. This makes it less likely that the downlink PRB usage reaches the threshold for load-based adaptive downlink scheduling to take effect. Consequently, there are fewer opportunities for load-based adaptive downlink scheduling to produce its intended benefits. If the downlink PRB usage reaches the threshold for load-based adaptive downlink scheduling to take effect, load-based adaptive downlink scheduling can take effect, but the benefits of downlink experience satisfaction guarantee may decrease.

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

Table 6-5 describes the base station model requirements of high-resolution MU pairing and adaptive MU resource management.

Table 6-5 Base station model requirements of high-resolution MU pairing and adaptive MU resource management

Function	Sub-Function	Base Station Model
High-resolution MU pairing	Downlink moving UE pairing optimization	3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
	MU-MIMO spectral efficiency optimization	
	PDCCH multi-dimension joint scheduling	
	Others	3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910. - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.
Adaptive MU resource management	Interference rejection precoding for downlink MU-MIMO in NLOS scenarios	3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
	Others	3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910. - DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units can be used. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR TDD-capable low-frequency RF modules with 32T32R and above can be used. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

6.3.4 Cells

AHR Capacity Upgrade 2.0 requires the maximum number of PDSCH layers to be set to eight or more. That is, the **NRDUCellPdsch.MaxMimoLayerNum** parameter must be set to a value greater than or equal to **LAYER_8**. In addition, the cell requirements listed in [Table 6-6](#) must be satisfied.

Table 6-6 Cell requirements of high-resolution MU pairing and precise AMC

Function	Sub-Function	Cell Requirement
High-resolution MU pairing	PDCCH multi-dimension joint scheduling	- The NRDUCellPdcchAl-go.PdcchMaxAggLevel parameter is set to AGG_LVL_8_16_ADAPT. - The NRDUCellPdcchAl-go.PdcchEdgePwrCtrlPolicy parameter is set to BASED_ON_CSI_RPT.
	MU pairing enhancement	When the NRDUCell.LampSiteCellFlag parameter is set to YES , the RRU.RT parameter cannot be set to EPRRU in the RRU MO corresponding to the SECTOREQM.ANT1CN , SECTOREQM.ANT1SRN , and SECTOREQM.ANT1SN parameters.
	Others	None
Precise AMC	Interference rejection precoding for downlink MU-MIMO in NLOS scenarios	The NRDUCell.SlotAssignment parameter cannot be set to 2_3_DSUUU .

Function	Sub-Function	Cell Requirement
	DMRS pseudorandom optimization	• The NRDUCell.NrDuCellNetworking-Mode parameter is set to NORMAL_CELL or FUSION_CELL. • NRDUCellTrp MOs have been configured for the cell, and the NRDUCellTrp.TrpType parameter is set to DEFAULT in all of these MOs. • The RRC_RECFCG_UL_SCH_OPT_SW option of the NRDUCellUISch.UlSchAlgoSwitch parameter is selected.
	Others	None

6.3.5 Others

None

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

Table 6-7 describes the parameters used for function activation.

Table 6-7 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
AHR Switch	NRDUCellFeatureSw.AhrSwitch	AHR_CAPC_UPGRADE_PHASE2_SW	Select this option when functions related to AHR Capacity Upgrade 2.0 need to be enabled.
MU-MIMO Optimization Switch	NRDUCellAlgoSwitch.MuMimoOptSwitch	None	Turn on this switch when high-resolution MU pairing needs to be enabled.
DL SRS MU-MIMO Freq Selection Threshold	NRDUCellPdsch.DLSrsMuMimoFreqSelThld	None	Use the recommended value when high-resolution MU pairing needs to be enabled.

Parameter Name	Parameter ID	Option	Setting Notes
DL SRS MU-MIMO PreSnr Threshold	NRDUCellPdsch.DLSrsMuMimoPreSnrThld	None	Use the recommended value when high-resolution MU pairing needs to be enabled.
DL Gathering MU-MIMO Optimization Switch	NRDUCellPdsch.DLMuGatherOptSw	None	Turn on this switch when high-resolution MU pairing needs to be enabled.
DL MU-MIMO Scheduling Supplement Switch	NRDUCellDLMimo.DLMuMimoSchSupplementSw	- SPECTRAL_EFFICIENCY_OPT_SW - DL_MOBILE_USER_PAIR_SW - MU_MIMO_USER_SELECT_OPT_SW - DL_PRECISE_GROUP_OPT_SW	It is recommended that all these options be selected when high-resolution MU pairing needs to be enabled.
MU-MIMO Switch	NRDUCellAlgoSwitch.MuMimoSwitch	PDCCH_MULTICAST_DIM_JOINT_SCHED_SW	It is recommended that this option be selected when high-resolution MU pairing needs to be enabled.
MU-MIMO Switch	NRDUCellAlgoSwitch.MuMimoSwitch	- DL_MU_RBG_PWR_NORM_ADAPT_SW - DL_MU_NLOS_IR_PRECODING_SW	It is recommended that these options be selected when adaptive MU resource management needs to be enabled.
SRS PreSINR Threshold	NRDUCellPdsch.SrsPresnrThld	None	Use the recommended value.
DL SRS MU-MIMO Rank Mode	NRDUCellDLRank.DLSrsMuMimoRankMode	None	It is recommended that this parameter be set to ENHANCE_MU_RANK when adaptive MU resource management needs to be enabled.

Parameter Name	Parameter ID	Option	Setting Notes
MU-MIMO Rank Enh Eqv SINR Thld	NRDUCellDirank.MuMimoRankEnhEqvSInrThld	None	Use the recommended value.
DL MU Intrf Reject Precoding Policy	NRDUCellPdschPrecode.DLMuIRPrecodePol	None	It is recommended that this parameter be set to MU_ALLUSER_POWER_ENH when adaptive MU resource management needs to be enabled.
DL DMRS Switch	NRDUCellDmrs.DlDmrsSwitch	DL_DMRS_SC RAMBLE_POLICY_SW	Select this option when adaptive MU resource management needs to be enabled.
Downlink Schedule Policy	NRDUCellDisch.DlSchPolicy	None	Use the recommended value.

6.4.1.2 Using MML Commands

Before using MML commands, refer to [6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling AHR Capacity Upgrade 2.0
MOD NRDUCELLFEATURESW: NrDuCellId=0, AhrSwitch=AHR_CAPC_UPGRADE_PHASE2_SW-1;
//Enabling MU pairing enhancement
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, MuMimoOptSw=ON;
MOD NRDUCELLPDSCH: NrDuCellId=0, DIMuGatherOptSw=ON, DISrsMuMimoPreSnrThld=0, DISrsMuMimoFreqSelThld=10;
//Enabling MU-MIMO UE selection optimization, MU-MIMO spectral efficiency optimization, downlink moving UE pairing optimization, and precise grouping optimization
MOD NRDUCELLDLMIMO: NrDuCellId=0, DLMuMimoSchSupplementSw=SPECTRAL_EFFICIENCY_OPT_SW-1&DL_MOBILE_USER_PAIR_SW-1&MU_MIMO_USER_SELECT_OPT_SW-1&DL_PRECISE_GROUP_OPT_SW-1;
//Enabling PDCCH multi-dimension joint scheduling
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, MuMimoSwitch=PDCCH_MULTI_DIM_JOINT_SCH_SW-1;
//Enabling RBG-level power normalization adaptation for downlink MU-MIMO and interference rejection precoding for downlink MU-MIMO in NLOS scenarios
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, MuMimoSwitch=DL_MU_NLOS_IR_PRECODING_SW-1&DL_MU_RBG_PWR_NORM_ADAPT_SW-1;
MOD NRDUCELLPDSCHPRECODE: NrDuCellId=0, DIMuIRPrecodePol=MU_ALLUSER_POWER_ENH;
```

```
MOD NRDUCELLPDSCH: NrDuCellId=0, SrsPresInrThld=2;  
//Enabling enhanced MU rank  
MOD NRDUCELLDLRANK: NrDuCellId=0, DISrsMuMimoRankMode=ENHANCE_MU_RANK,  
MuMimoRankEnhEqvSinrThld=75;  
//Enabling DMRS pseudorandom optimization  
MOD NRDUCELLDMRS: NrDuCellId=0, DIDmrsSwitch=DL_DMRS_SCRAMBLE_POLICY_SW-1;  
//(Optional) Enabling load-based adaptive downlink scheduling  
MOD NRDUCELLDLSCH: NrDuCellId=0, DISchPolicy=EPF;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling AHR Capacity Upgrade 2.0  
MOD NRDUCELLFEATURESW: NrDuCellId=0, AhrSwitch=AHR_CAPC_UPGRADE_PHASE2_SW-0;
```

6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

If the total number of PRBs paired for downlink MIMO (**N.ChMeas.MIMO.DL.Pair.PRB**) increases in heavy-load scenarios, high-resolution MU pairing has taken effect.

After adaptive MU resource management is enabled, this function has taken effect if the proportions of different ranks in a cell change and the number of RRC reconfiguration messages increases within a few minutes. In this case, the value of the *scramblingID0* or *scramblingID1* IE in the RRC reconfiguration messages changes from the PCI of the serving cell to the C-RNTIs of UEs.

6.4.3 Network Monitoring

The following indicators can be used to observe the average downlink UE throughput and average downlink cell throughput, respectively:

- **User Downlink Average Throughput (DU)**
- **Cell Downlink Average Throughput (DU)**

The network monitoring of this feature also involves other indicators whose definitions are described in [Table 6-8](#).

Table 6-8 Definitions of indicators related to AHR Capacity Upgrade 2.0

Indicator	Definition
Average downlink MCS index	$\frac{\text{SUM}(k \times \text{N.ChMeas.PDSCH.MCS}.k)}{\text{SUM}(\text{N.ChMeas.PDSCH.MCS}.k)}$ $k = 0 \text{ to } 28$

Indicator	Definition
Downlink PRB usage	$N.PR.B.DL.Used.Avg/N.PR.B.DL.Avail.Avg$
Downlink IBLER	$(N.DL.SCH.QPSK.ErrTB.Ibler + N.DL.SCH.16QAM.ErrTB.Ibler + N.DL.SCH.64QAM.ErrTB.Ibler + N.DL.SCH.256QAM.ErrTB.Ibler)/(N.DL.SCH.QPSK.TB + N.DL.SCH.16QAM.TB + N.DL.SCH.64QAM.TB + N.DL.SCH.256QAM.TB)$
Number of PRBs for downlink MU pairing	$N.ChMeas.MIMO.DL.Pair.PR.B$
Uplink PRB usage	$N.PR.B.UL.Used.Avg/N.PR.B.UL.Avail.Avg$
Uplink IBLER	$(N.UL.SCH.HalfPiBPSK.ErrTB.Ibler + N.UL.SCH.QPSK.ErrTB.Ibler + N.UL.SCH.16QAM.ErrTB.Ibler + N.UL.SCH.64QAM.ErrTB.Ibler + N.UL.SCH.256QAM.ErrTB.Ibler)/(N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)$
Uplink RBLER	$(N.UL.SCH.HalfPiBPSK.ErrTB.Rbler + N.UL.SCH.QPSK.ErrTB.Rbler + N.UL.SCH.16QAM.ErrTB.Rbler + N.UL.SCH.64QAM.ErrTB.Rbler + N.UL.SCH.256QAM.ErrTB.Rbler)/(N.UL.SCH.HalfPiBPSK.TB + N.UL.SCH.QPSK.TB + N.UL.SCH.16QAM.TB + N.UL.SCH.64QAM.TB + N.UL.SCH.256QAM.TB)$
UE access success rate	$N.RRC.SetupReq.Succ/N.RRC.SetupReq.Att$
First packet delay at the RLC layer	$N.Traffic.DL.RlcFirstPktDelay.Time/N.Qos.DL.RlcFirstPktDelay.Num$

7 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

8 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

9 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

10 Reference Documents

1. *Channel Management*
2. *MIMO (TDD)*
3. *FWA*
4. *High Speed Mobility*
5. *Hyper Cell*
6. *Cell Combination*
7. *Distributed Antenna Solutions (Low-Frequency TDD)*
8. *Carrier Aggregation*
9. *CA Experience Improvement*
10. *Interference Avoidance*
11. *Massive MIMO iBeam (Low-Frequency TDD)*
12. *Modulation Schemes*
13. *Multi-Frequency Smart Aggregation*
14. *Massive MIMO AHR (Low-Frequency TDD)*
15. *Power Control*
16. *Positioning*
17. *Standards Compliance*
18. *License Control Item Lists*
19. *Massive MIMO iBeam (Low-Frequency TDD)*
20. *LTE and NR Zero Bufferzone*
21. 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"